Sahaj Bhadja

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Summary

Software Engineer with 2+ years building **AI backend systems** and **APIs** in Python, Java, and TypeScript, including multi-provider LLM routing, model inference services, authentication, logging, and cloud deployments, from requirements through production.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Advanced NLP, Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Data Retrieval, Software Engineering, Object-Oriented Programming, Operating Systems

Experience

USC Dornsife **September 2025 – Present**
Forward Deployed Engineer *Los Angeles, CA*

- Restructured fault-tolerant asynchronous **LLM orchestration** with FastAPI workers, explicit job-state transitions, and failure isolation, processing 1K+ AI jobs at 3x throughput without cascading failures under concurrent load.
- Designed multi-tenant storage on Azure SQL with **row-level security** for 500+ users, cutting dashboard latency from 1.2s to 350ms, and shipped integrations through CI/CD at 99%+ reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a **unified model layer** that routes requests across OpenAI, Anthropic, and Qwen models with dynamic switching, vision fallbacks, and timeout handling, serving 10K+ daily requests on AWS.
- Logged model name, feature, token counts, and estimated **cost per call** in MySQL with static pricing configuration, enabling cost-aware model usage while cutting feedback turnaround 60% for 150+ instructors.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *Los Angeles, CA*

- Built a GPT-based Decision Transformer in **PyTorch** for offline reinforcement learning, with data pipelines over 100K+ trajectory logs, improving long-horizon trajectory planning 18% in simulation through reproducible experiments.
- Designed a multi-head **attention** encoder fusing map geometry with agent trajectories, benchmarking it against three baselines with Weights & Biases and improving closed-loop planning stability 15% across simulation evaluations.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Converted stakeholder **requirements** into scoped tasks with Microsoft and GitHub teams, building cross-platform microservices backed by 20+ automated unit and integration test suites in nightly Jenkins pipelines.
- Optimized large-scale **API integrations** with pagination, field filtering, and caching, improving response times 30% and redesigning AWS Glue and S3 pipelines for 68% higher efficiency across production workloads.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built TypeScript Node.js microservices with JWT **authentication** and role-based **authorization** for customer, billing, and lead workflows, syncing 2M+ records across locations through real-time WebSocket updates securely and reliably.
- Added **logging** and monitoring with Sentry, Azure Monitor, and Kibana, and shipped through Jenkins CI/CD with automated tests and Docker staging releases, cutting deployment time 50% across production releases.

Skills

Languages: Python, Java, TypeScript, JavaScript, SQL, C++
AI & ML: LLM APIs, GPT-4o, Claude, Model Routing, Prompt Engineering, RAG, Embeddings, FAISS, PyTorch, Transformers
Backend & APIs: FastAPI, Flask, Spring Boot, Node.js, REST APIs, Microservices, JWT, Role-Based Access Control
Data & Cloud: PostgreSQL, Azure SQL, MySQL, AWS, Azure, GCP, Docker, Kubernetes
Operations: Logging, Sentry, Kibana, Azure Monitor, CI/CD, Jenkins, Git, PyTest

Projects

Multi-Model API Gateway | *TypeScript, Node.js, Redis, PostgreSQL, Docker*

- Designed a **unified API gateway** that exposes several LLM providers through one OpenAI-compatible endpoint, with per-key authentication, rate limits, usage metering, request logs, **automatic failover**, and observability.

Self-Hosted Model Inference Service | *Python, PyTorch, FastAPI, Kubernetes, Docker*

- Designed a **model serving** service that runs an open-weight LLM through PyTorch workers, streams tokens over FastAPI, autoscales on queue depth, and tracks **latency and throughput** metrics.

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, Clerk, PostgreSQL, GCP*

- Built an **AI application** with LLaMA-2, transformer embeddings, and FAISS semantic search over 50K+ documents with Clerk authentication, cutting retrieval latency 40% at 95% query accuracy in production.